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Teaching across modalities: Examining instructor presence in HyFlex environments

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ABSTRACT

HyFlex learning experiences are increasingly being offered to accommodate disruptions to learning experiences and offer flexibility to instructors and students. Limited research has examined meaningful instructor presence in HyFlex learning environments. Using the Community of Inquiry (CoI) Framework, the purpose of this case study-mixed methods research was to explore how instructors established presence in HyFlex learning environments. The instructional approaches and perspectives of three instructors teaching the same introductory undergraduate design-thinking course and student attendance and responses to the CoI survey were compared across course sections. Results highlight several ways that instructors established and maintained their presence in HyFlex environments. This study offers insight into the design and facilitation of HyFlex learning.

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Introduction

Research suggests the ways in which instructors facilitate learning and establish presence substantially influence students' learning, sense of community, participation, and engagement (Adnan, 2017; Polat, 2022; Richardson et al., 2015). While instructor presence has been studied in both face-to-face and online learning environments (Glazier & Harris, 2021), less is known about how instructor presence is formed in blended learning environments, that combine face-to-face and online elements. As educators increasingly adopt flexible and accessible blended learning approaches to meet diverse and evolving student needs (Mentzer et al., 2024), understanding how instructor presence is created across modalities is necessary for the effective design and facilitation of blended learning.

Furthermore, since COVID-19, hybrid flexible (HyFlex) learning environments are increasingly being implemented (Mentzer et al., 2024; Wong et al., 2023)—blended environments offering students choice as to how they attend each class session, while requiring instructors to manage multiple modalities simultaneously for remote and face-to-face students (Beatty, 2019). Unlike other blended learning models, HyFlex instructors facilitate engagement and community across multiple participation modes concurrently (Beatty, 2019), with inconsistent instructor presence across modalities potentially negatively affecting student belonging and learning outcomes (Buckley et al., 2024). Previous research on HyFlex environments has predominantly considered learners' perspectives and experiences, without considering the instructor role in shaping these learning experiences (Magana et al., 2022). Given that higher education is predicted to remain a blend of

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in-person and online learning in the future (Mentzer et al., 2024; Sellnow-Richmond et al., 2020), examining HyFlex learning environments and how instructors establish presence, an important component of meaningful learning experiences, can inform improvements in how HyFlex learning is facilitated across modalities. The following research questions inspired our inquiry:

1. How do HyFlex instructors incorporate presence in their courses?
2. How do differences in instructor presence in HyFlex learning environments impact student perceptions and attendance?

Literature review

HyFlex learning

Although not a new approach, HyFlex learning was implemented extensively to address challenges created by the COVID-19 pandemic (e.g., accommodating sick and quarantined students and instructors) (Kohnke & Moorhouse, 2021; Mentzer et al., 2024), offering educators ways to accommodate diverse and changing student needs and learning contexts through flexibility and autonomy in the learning process (Nunes et al., 2023) and giving students choice between attending classes face-to-face or online, synchronously or asynchronously (Beatty, 2019). Some educators see HyFlex as a way to promote flexible attendance options for students and promote accessible education (Clifft & Assiouras, 2023).

HyFlex, a combination of the terms *Hybrid* and *Flexible*, is an instructional format that simultaneously combines both online and face-to-face learning environments (Beatty, 2019; Heilporn & Lakhal, 2021), offering students flexibility in how they choose to participate in a learning experience from class-to-class (Beatty, 2007). For instance, a course may be designed to live-stream activities, therefore allowing both face-to-face and remote students to listen to presentations and interact with peers synchronously regardless of the modality they select (Kohnke & Moorhouse, 2021). Adapting Beatty (2007)'s work, Mentzer and Mohandas (2022) created the Interactive Synchronous HyFlex model, which offers learner choice (students select attendance mode between face-to-face or online synchronous), equity (instruction is equivalent across modalities), reusability (course materials are equally available), and accessibility (all course aspects are equally accessible).

Previous HyFlex research has predominantly explored students' perspectives: analyzing students' autonomy, competence, and relatedness (Bozan et al., 2024; Mentzer et al., 2024, 2025); considering the impact of HyFlex approaches on students' course grades (Magana et al., 2022; Mentzer et al., 2024), and examining student challenges and perceptions (Buckley et al., 2024; Detyne & Koch, 2023; Eduljee et al., 2023). Fewer studies have considered instructors' perspectives within HyFlex learning environments (Amirova et al., 2023; Boehm & Boerboom, 2023), especially in problem-centered contexts, where instructors must balance facilitation, content expertise, and student collaboration across multiple modalities (Mohandas, 2022; Raes et al., 2020), as they create the foundation necessary for the learning experience, facilitate learning using appropriate just-in-time prompts and support, and offer closure at the end of the experience (Ertmer & Koehler, 2018). The challenges and complexity are amplified in HyFlex environments, where instructors must simultaneously manage face-to-face and remote students and technologies, while maintaining pedagogical effectiveness.

Community of inquiry and instructor presence

The Community of Inquiry (CoI) framework has been extensively used to offer instructors ways to consider and guide students' online learning and establish presence by characterizing productive online communities of inquiry through three elements (Shea et al., 2022): cognitive presence ("the extent to which the participants... are able to construct meaning through sustained

communication,” Garrison et al., 1999, p. 89), social presence (“the ability of participants... to project themselves... as ‘real’ people,” Garrison et al., 1999, p. 94), and teaching presence (“the design, facilitation, and direction of cognitive and social processes” for learning, Anderson et al., 2001, p. 5). In short, “the CoI framework summarizes the instructional, social, and cognitive processes through which productive online communities of inquiry are developed”—communities resulting from instructors creating interactive learning and intentionally guiding deep consideration of content and students who are active in their learning process (Shea et al., 2022, p. 148). While the CoI framework was created to consider online communities, recent research has established its usefulness in exploring HyFlex learning environments (see Adam et al., 2025; Buckley et al., 2024; Jones et al., 2024).

Instructor presence is conceptualized as the intersection of teaching presence and social presence, capturing instructors’ efforts to project themselves as real people as they use intentional strategies to structure, facilitate, and manage learning (Richardson et al., 2015). Teaching presence has three areas: design and organization (e.g., directing course navigation; sharing availability with students), facilitating discourse (e.g., encouraging participation, providing tips), and direct instruction (e.g., clarifying content, demonstrating how to do something) (Adam et al., 2025). Social presence also has three components: affective (e.g., sharing personal information, showing emotion), cohesive (e.g., using greetings, promoting collaboration), and open communication (e.g., acknowledging students’ contributions, asking questions to engage students) (Adam et al., 2025).

As instructors interact with students, they assume various roles (Richardson et al., 2015, 2016; see Table 1), influencing student participation (Jorge, 2010), learning facilitation (Joksimović et al., 2015), and student retention (Reio & Crim, 2013). In active learning environments, instructor roles shift from lecturer to facilitator, where presence can be emphasized through using tools (e.g., announcements, video) and actions (e.g., affirming students’ ideas, providing individualized feedback) (Richardson et al., 2015, 2016, 2024).

HyFlex instructor presence

Meaningful instructor presence can be understood as a holistic approach to student support that combines both emotional and academic elements (Robinson et al., 2017; Velasquez et al., 2013) and involves the creation of a structured yet nurturing learning with consistent communication, individualized support, and responsive dialogue with students. Instructor presence emerges through authentic interactions, careful attention to student needs, and monitoring of both academic progress and well-being (Robinson et al., 2017; Velasquez et al., 2013). As HyFlex environments combine online and face-to-face elements, for instructors to establish effective presence, they must do so across modalities.

Instructor presence is an essential aspect of effective learning environments, as how instructors facilitate learning experiences, support and communicate with students, provide direction for course activities, and establish themselves as real people can influence student retention, actual and perceived learning, and satisfaction with a course (Glazier & Harris, 2021; Richardson et al.,

Table 1. Descriptions of roles used to establish presence.

Instructor role	Description
Advocating	Supporting learning by offering encouragement, tips, and availability
Facilitating	Shaping discourse through engaging prompting and question asking and summarizing shared ideas
Sensemaking	Offering understanding through clarification, demonstration, resources, feedback, and modeling
Organizing	Structuring the course through sharing course information, expectations, requirements, important dates, and guidance for completing course requirements
Maintaining	Keeping course flow through guiding navigation, technology use and platforms, reminders, and feedback requests.

2015; Rosser-Majors et al., 2022; Sheridan & Kelly, 2010; Vesely et al., 2007). For instance, Glazier and Harris (2021) surveyed over 2,000 students asking them about their best courses, across face-to-face and online experiences. Regardless of format, students indicated that clearly articulated instructions and expectations; interactivity; instructor availability and promptness in responses; class organization; consistent deadlines; interesting discussions; and engaged, caring, knowledgeable instructors were most important. All of these items align closely with instructor presence, as conceptualized using the CoI framework. Other research has considered relationships between instructor presence and technology used during synchronous sessions, finding that technology use does not automatically lead to positive instructor presence (Martin et al., 2018). These findings underscore the nuanced and context-specific nature of instructor presence in HyFlex learning environments.

Methods

Research design

We used a case study-mixed methods (Guetterman & Fetters, 2018) approach to explore three instructors' HyFlex facilitation, as we sought to discover a potential range of generality in the presence established by instructors and to provide "explanatory power and greater generalizability than a single case study can deliver" (Miles & Huberman, p. 151). Each instructor and their class were considered an individual case, with data triangulated from four sources: instructor interviews, observations of recorded instructor facilitation, student responses to the CoI survey, and student attendance records. First, we conducted a within-case analysis to gain a sense of individual instructor's presence, their rationale for the specific approaches, and students' perceptions of their presence (Research Question 1). Next, by completing a cross-case analysis, we compared instructor presence across the cases to uncover similarities and differences in approaches and how this impacted student responses to the CoI survey and attendance (Research Question 2) (Merriam & Tisdell, 2016).

Context and participants

Participants included three instructors that taught an introductory undergraduate design-thinking course required for technology majors at a large U.S. midwestern public university and their students during fall 2022. Using a flipped classroom approach (see Tawfik & Lilly, 2015), the course prompted learners to consider materials and videos prior to meeting and complete hands-on activities during class time. Students completed three projects focused on building their design thinking processes. The course was broken into sections of approximately 35–40 students, with one instructor teaching a section. While all sections of the course followed the same schedule and completed the same assignments, instructors used different approaches for covering content. Instructors' experience with teaching the course ranged from 2 to 7 times. See Table 2 for participant demographic information.

The course used the Interactive HyFlex model to offer interactive, hands-on learning in a flexible way to students across two modalities: face-to-face or remote synchronous (Mentzer

Table 2. Participant demographic information.

	Tom	Ben	Lia
Age	26–40	26–40	26–40
Teaching Experience in Higher Education	6–10 years	2–3 years	0–1 years
Teaching Experience in HyFlex Format	1–2 years	1–2 years	0–1 years
Professional Background	K–12 Teacher	Engineer	Engineer
Number of Students	38	37	38

et al., 2024). Grounded in the CoI framework, the Interactive HyFlex model allows students to switch between modalities from one class to the next. While students were encouraged to attend the class in person, they could also attend remotely through Microsoft Teams during live class sessions, interacting with their instructor and teammates. In this model, course resources and information were shared using the learning management system (LMS), and Microsoft Teams was used to accommodate remote attendance, facilitate and record course sessions, and facilitate teamwork. All students were expected to use these tools to work with peers across modalities and to actively participate regardless of modality.

Data collection and analysis

We considered four primary data sources: 1) semi-structured interviews with instructors regarding how they established presence, 2) observations of recorded instructor facilitation of HyFlex class sessions throughout the semester, 3) student responses to the CoI survey, and 4) student attendance records. We used the indicators of an established framework (see Richardson et al., 2015) to develop the interview protocol and deductively analyze interviews and observed instructor facilitation, focusing on teaching presence (design and organization, direct instruction, facilitating discussion, and assessment) and social presence (affective, cohesive, interactive). The questions were piloted with a former course instructor and revised for clarity. Instructor interviews were conducted during the first half of the fall 2022 semester. Questions focused on prompting instructors to share specific strategies they used to establish instructor presence (see [Appendix A](#) for interview protocol).

At semester completion, we examined 15 recorded class sessions per instructor and class chats from Microsoft Teams for each instructor. We analyzed observations and interview transcripts to capture instructor presence. While the goal was not to achieve an exhaustive analysis, this approach allowed us to identify examples of how instructors created presence, their justification for the approaches they used, and differences across instructors. See [Appendix B](#) for sample coded examples. Two members of the research team independently coded interviews and observations and then compared coding and reached consensus through discussion. These data sources were used to answer the first research question.

To answer the second research question, students in each section of the course were asked to complete the CoI survey at the end of the semester (Arbaugh et al., 2008). Students' responses to items related to teaching presence and social presence were considered, as these two areas were used to define instructor presence. We calculated descriptive statistics of student responses and used a one-way ANOVA to compare students' responses across the three course sections. Additionally, we calculated average attendance modes for each instructor. At the beginning of each class session, undergraduate teaching assistants recorded whether students were present and whether they were face-to-face or remote. We created individual portraits for each instructor and then compared instructors' experiences to identify patterns within and across cases (Yin, 2018). This study was approved by the university Institutional Review Board, informed consent was obtained prior to collecting data, and pseudonyms were used to protect instructor identity.

Results

For each instructor, a case study was developed by identifying examples of teaching presence and social presence from observed facilitation, while using instructor interview responses to offer rationale for the facilitation choices being made. An overview of each case is presented to answer the first research question, followed by a cross-case analysis that compares student perceptions and attendance to answer the second research question. See [Appendix B](#) for samples of coded observations and interview transcripts. As a note, the instructors mentioned using Google Jamboard, a collaborative, digital whiteboard which is no longer available.

Case 1: Ben

During fall 2022, Ben was teaching the design thinking course for the seventh time and the fourth time in a HyFlex format. Overall, Ben valued the HyFlex approach to support course structure and used his professional experiences to communicate relevance: “I feel like one of the biggest goals of this course is to give them a group experience...to learn how to work better in a team,” and “we’re living in a world where you need to be able to connect with anyone, work with anyone, no matter whether they’re in the room with you or not.” Ben emphasized these beliefs with students during the first day of class when sharing examples of his professional work experience as an engineer and seeking credibility: “I had a lot of experience working with remote members, which is something this class will focus on a lot.”

For each class session, Ben used a system for tracking participation that helped monitor remote (commenting in chat) and face-to-face (writing on whiteboard) attendance. He kept session overviews concise, typically five to ten minutes. Using a variety of technologies (e.g., PowerPoint, Google Jamboard, Teams chat, screen sharing, videos) was a key aspect of how Ben established and extended his presence. Information flowed between him and students and across modalities, as he asked students to consider questions related to design in real-world situations. Students were given most of class to work collaboratively, with Ben checking in with groups to support their work (see Figure 1).

Although Ben indicated to his students that face-to-face was the best option for attending class, he believed that students had many potentially valid reasons for attending remotely, as “they wouldn’t have been in that class otherwise.” Before student presentations at the end of the semester, he discussed attendance in this way:

For the week of presentations, I would like everyone to be here in person. I think that will give you the easiest chance to show you’re actually invested in the project, at least for the last couple of days. That would give me a better idea just how important it is to you guys to show up and do well for this class...In general, I think you guys have been really responsible with remote options. So, just like with the rest of the class, if something comes up morning of, and you need to present remotely, we will have the option to do that. But please try your best to be in class for your presentation...

All students, regardless of whether attending remotely or face-to-face, were required to log into Microsoft Teams at the beginning of each class. Ben believed that this benefited all students, noting that he tells students, “if you want to see my screen you probably will want to connect.” This setup allowed Ben to create private channels within Microsoft Teams for each team to have their own accessible workspace, and he would pop into these to check-in on students. Additionally,

Hello! As you log into Teams...

Class Slides available in Teams in the General Files

- Please send a **message in Chat** if you are remote or **write your name on the white board**.
- Let me know if you are having trouble with teams
- Read over the discussion question responses in **Discussions Channel** on Teams
 - Give a reaction to responses that gave a good Design Thinking examples you didn’t know about.

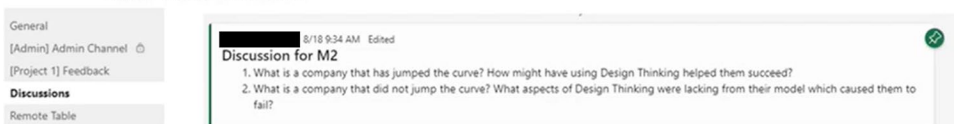


Figure 1. Example of Ben’s HyFlex setup.

his technology use supported extensive availability both within (e.g., “Let me connect to my phone so we can actually talk a little bit better.”) and outside (e.g., “I just had a meeting with them in some random parking lot” *via* his phone) of the classroom. The CoI survey responses from Ben’s students indicated positive perceptions of social presence and teaching presence.

Ben’s social presence

Across observed sessions and modalities, Ben consistently greeted students, used their names, offered approval (e.g., “You guys rock!”), and shared his values related to teamwork and using technology to facilitate collaboration (e.g., encouraging students to use “software to work with people” and foster meaningful communication). At the beginning of each class, he shared his screen *via* Teams with a message that brought remote students into the conversation (“Say hi if remote!”). Then, he acknowledged and greeted each remote student, using their name, before checking for questions: “Hey Reece! Hey Dana! Hey Matt! So, if you have any questions, please let me know.” Throughout a class meeting, Ben was observed to consistently share his screen through Teams, encourage the use of technology to facilitate collaboration (e.g., brainstorming through Google Jamboard), and use group language (e.g., “We’re doing project one today... we’re going to try to get into it as soon as we can.”).

As Ben facilitated group work, he simultaneously managed students face-to-face and remotely, acknowledging students’ ideas and effort (“Special shoutout to Brandon—Brandon noted that the one from the reading didn’t contain any constraints.”). When students were forming groups, Ben required students to use a team contract and carefully consider modality: “How do you want to communicate outside of class? How do you feel about remote meetings? How do you feel about in-person meetings?” In one instance, where students were working in pairs, he directed face-to-face students to “move over to the table over here in the back,” while directing remote students to “stay in the chat.” Inviting responses from students was a key aspect of Ben’s presence, as he was observed multiple times asking students to respond to questions both verbally or through the chat, promoting interaction and connections with the content and offering a different perspective. In one instance, he responded in this way: “I think there’s another cool aspect of design thinking though in that example... So, empathy was one of those steps from the homework... what was something we did this week?”

Ben’s teaching presence

Ben’s teaching presence consisted of sharing resources (e.g., “Google Scholar is another option that’s just as easy to use.”), offering frequent reminders (e.g., “There’s nothing due for Thursday before class”), and constantly offering direction on how to successfully navigate class meetings (e.g., “So go ahead... I’ll tag everyone in Teams, put feedback and if you guys have questions.”). Moreover, Ben spent much effort modeling how to use Teams to facilitate collaboration and communication, requiring students to schedule a remote meeting for 30 min in place of class; creating meetings within their course Teams channel using speaker microphones on either side of the classroom to facilitate a group feedback session; and proactively prepping a Teams message that would go out to student teams (see [Figure 2](#)).

Technology played a central role in the teaching presence Ben created. For instance, rather than assigning topics for a project in class, he created a Google Doc, facilitating easy sharing across modalities. Finally, Ben was seen frequently checking in with learners, assisting with troubleshooting technology, and quickly adjusted methods to accommodate remote learners’ needs. In one instance, he helped a remote student pair up with a face-to-face partner and configure technology to collaborate. In another instance, a remote student asked about a grade, which Ben agreed to investigate and get back to the student. Interestingly, Ben believed that most of his feedback was given through the LMS and that he “didn’t go into Teams much.” However, in the recorded class sessions, Ben was observed summarizing ideas shared from team

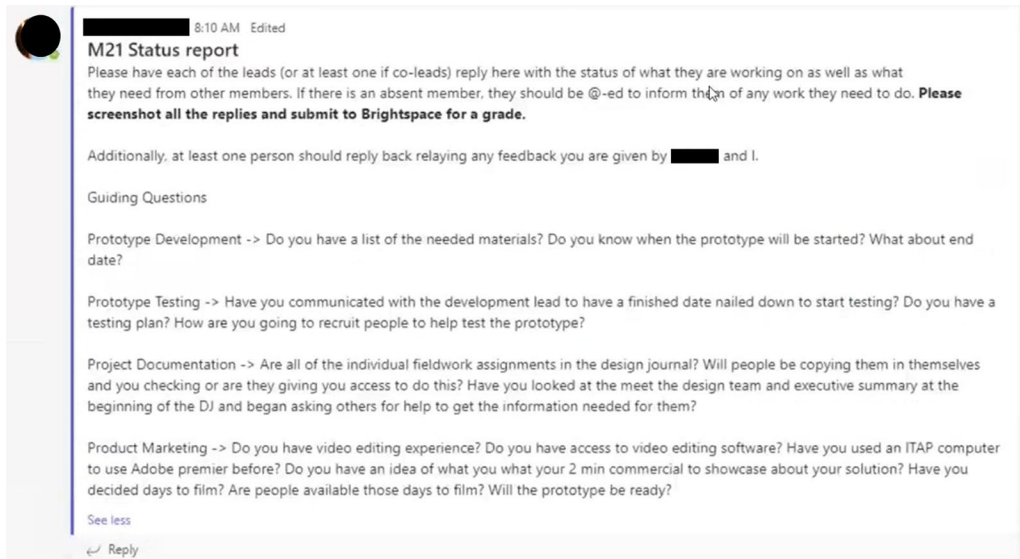


Figure 2. Example of Ben Modeling Microsoft Teams Use.

chat conversations and providing both face-to-face and online students with formative feedback on their work and discussions.

Case 2: Lia

Lia was teaching the course for the second time during fall 2022, which was also her second time teaching in a HyFlex format. She explained the use of HyFlex to her students “as a short-term option... If students need to be remote it’s a way for them to stay up to date with what we’re working on, but it’s not an alternative [to attending class physically].” She added that because the class is project-focused, “it is important that you’re in class... because you have other teammates who are relying on you.” During the first class, Lia explained attendance to students in this way: “If there is a legitimate, important reason why you can’t be here face-to-face, you can join us remotely. We would prefer that you were here face-to-face though. With the design thinking class, there’s a lot of hands-on work, there’s a lot of group work, and being here in person, face-to-face makes that go a lot more smoothly. We do understand though that there are certain situations where you can’t be here, and we want to make sure that you don’t lose out on being a part of class.” Prior to the culminating project presentations, Lia communicated the attendance in this way to her students:

You need to be in class for both Wednesday and Monday, unless you have an excused absence. Being absent on one of those days will cause you to get a zero on your presentation, which I don’t want to do. I want to make sure you get credit for the work that you do, but we also want you to come to class and participate with q and a. So, it’s really important that you come to class both days.

For class sessions, Lia typically provided an overview at the beginning, lasting between 10 to 20 min. She consistently used PowerPoint presentations to guide sessions and was observed sharing her screen sometimes as she navigated the LMS or other documents. However, in a couple of observations, she experienced challenges with screen sharing and managing multiple technologies (e.g., presentation showing on large screens in room, but not on the recording). Lia often used part of class sessions to facilitate class discussions—sometimes prompting students to first respond in a Teams chat and then build on their responses through a synchronous verbal conversation. As Lia facilitated each session, she was simultaneously considering the experiences of face-to-face and blended students and checking on them: “Do you have access? If you don’t

have access, what can I do? They were either responding on the chat ... or they were responding verbally through my audio.”

Lia required all students to log into Teams at the beginning of each class, regardless of whether they were attending face-to-face or remotely and used several technologies to facilitate learning across modalities (e.g., Teams chat, Google Jamboard, screen sharing). By using these tools, she was simultaneously enhancing learning for both face-to-face and remote students (through screen sharing, all students “have a good view of what I’m sharing on my screen”) and offering ways of extending contact with content (by using the Teams chat tool, “it helps the students to be able to kind of record their thoughts,” facilitating organization and preservation). The CoI survey responses from Lia’s students indicated positive perceptions of social presence and teaching presence.

Lia’s social presence

During observed class sessions, Lia greeted learners (e.g., “Welcome to another exciting day of Tech 120!”), asking remote students to type their name in the Teams chat and face-to-face students to write their name on the white board. In multiple instances, Lia was observed referring to remote participants as “online friends.” While she commonly chatted with students informally at the beginning of class (e.g., asking what everyone was dressing up as for Halloween), limited strategies were implemented to fully integrate remote students. Specifically, she was observed checking with remote students about general things but not asking remote students if they had anything to add in the class discussion she often facilitated. However, when discussing course content, she consistently used PowerPoint slides to promote understanding and sometimes shared her screen. She consistently used group-friendly language (e.g., “...surveys that we’re gonna be doing for Project 3, just like we did for Project 2...”) and used students’ names. To support group collaboration across modalities, she offered guidance on both using the software (e.g., “Make sure you started the meeting.”) and hardware (e.g., “Would you like a speaker puck?”).

Lia was observed in most sessions checking with students across modalities to make sure they were progressing through the class learning experience effectively (e.g., “Can you hear me OK?”, “Do you have any questions on what this is, what this means?”) and offered approval for students’ work (“Wow! So exciting! So much chatter and excitement from such a great solution!”). Creating an inclusive, “safe environment where everyone felt comfortable and respected” also appeared important to Lia, as she specifically considered her class as a “community,” offered availability (“...you can always reach out to me *via* e-mail or Teams message”), and checked in with students who attended remotely two times in a row.

Overall, using Teams was a major factor in Lia’s social presence, as she used this to display her video, created private channels for teams to have “their space,” reminded students to have their cameras on, facilitated collaboration through it (e.g., “You can go to a chat in Teams and drop your PDF in there and exchange them that way...”), and offer support.

Lia’s teaching presence

Lia often offered reminders to students (“Before our next meeting, you will submit this class work that you’re doing here.”), emphasized that everyone should be logged into Teams as it is the “accepted platform for our class,” provided guidance with how to navigate a class meeting (e.g., “Give feedback to the group that comes after your video.”), and offered feedback on students’ ideas and work (e.g., “Do you have access to your user group? If you want your user to be kindergarteners, do you have access to a group of kindergarteners?”). Feedback was commonly solicited using the Teams chat tool: “I have individual channels. For every one of the teams, so I can drop a message into each one of those channels to say, ‘Hey reply to this and let me and your teammates know where you are so far on this assignment.’” In another instance, Lia used a system displayed *via* PowerPoint to help prioritize interactions with students (See [Figure 3](#)).

Facilitating discourse was a central component of Lia's approach, as she commonly prompted students to consider content deeply, summarized shared ideas, and offered tips. While she was observed using Teams consistently to share content with students, in these observations, the content was not equally available to remote students or absent students. For instance, during one class meeting, she played a video, where students were observed watching the video on the large screen in the classroom, while others viewed it from their computer screens. Prior to playing the video, she muted Teams, allowing only face-to-face students to hear the audio, and in the recorded meeting, audio for the video cannot be heard. Although she had previously checked there were no remote students, this is a limitation for classroom management and potentially supporting absent student needs.

Case 3: Tom

When the study took place, Tom was teaching the course for the fifth time. When starting a new semester, Tom described using the same steps each time to introduce the HyFlex format to students: 1) "I send probably the exact same announcement about the core structure," 2) point out information in the syllabus to communicate "what my expectations are" with the "remote attendance option," and 3) have the students "practice interacting with one another using Microsoft Teams" during the first class. Additionally, he explained delaying talking about the "nuts and bolts" of HyFlex until the second class, as "HyFlex is not the point of the class."

He was observed facilitating each class session in a similar manner: First, checking in with students or chatting conversational, particularly focused on the face-to-face students. For instance, sometimes during this informal time, he looked around the room, prompting responses from face-to-face students (not extending the same prompt to remote students) or talking to face-to-face students with his microphone muted, preventing remote students from hearing or joining in the conversation. Next, he used the first part of class (typically 15–25 min, however in some instances he talked for the entire class session) sharing an overview of course content and assignments. During these overviews, while streaming video from a webcam, information flowed

WRITE A NUMBER ON YOUR WHITEBOARD THAT CORRESPONDS TO YOUR GROUP'S **CONFIDENCE LEVEL RIGHT NOW:**

 **3** We are feeling great!

 **2** We are doing OK.

 **1** Help!

"As a group choose a number to write on your whiteboard, one through three, where three is 'Yea we've got this. Don't even bother us. We're rocking and rolling.' And, I won't even come bother your group. Two is, 'We're okay, maybe we could use a little help, but we probably got it.' And, one is, 'Please help. We need support right now.' I will prioritize my ones, help my twos when I have time, and let my threes just work through what they got."

Figure 3. A Feedback Strategy Used by Lia.

unidirectionally. That is, Tom used a lecture-style approach that involved talking to students, sharing his screen (e.g., LMS navigation, assignment details), while often writing on or referring to content on a dry erase board in the classroom. Content on the board was usually not readable in the video recordings. He opted to not use PowerPoint slides, as he viewed this as “not really necessary to have redundant information on like a PowerPoint slide.” The remainder of class time was usually for students to work collaboratively, where Tom walked about the room, checking in with groups. During this time, on multiple instances, Tom was observed encouraging students to use the white board for collaboration: “Write it up on the white boards or you could certainly use Google Slides or PowerPoint or something like that to make a table. You could do either one. Just know that if you put it on your computer, I’m going to have to stare awkwardly over your shoulder, and maybe that’s uncomfortable for you, maybe it’s not.” (See Figure 4 for typical class setup).

During his interview, Tom stressed that remote attendance should be used only when “absolutely necessary” and shared “potential reasons why [students] could be working remotely.” This matched with observations of class recordings, where he often explained to students his expectations related to their attendance and participation. Prior to final presentations, Tom communicated the following to his students:

Unless you have a valid excuse to be remote, and by that I mean that you are covered through the medical emergency policy by the Office of the Dean of Students, you have COVID and need to quarantine, like those are two legitimate reasons why you would not be in attendance face-to-face... in that case, I would know about that beforehand... If you don’t come to class, if it’s unexcused, even if you feel that it’s a legitimate excuse, but it doesn’t fall under one of those policies I just talked about, you will get a 0 on your presentation, regardless if it’s your presentation day... Nothing is more awkward than receiving a zero on your presentation before you presented, and you will still be expected to present.

Although Tom only required students to log onto Teams if they had a teammate that was remote, he logged in and shared his screen each class session and felt comfortable with shifting as necessary to meet changing course composition: “If I happen to have anybody that is working remotely that day, I’m prepared to either put them all into a group or have them work individually.” With remote students in attendance, Tom reminded both remote students and their face-to-face teammates of how to navigate HyFlex collaboration. He also offered alternative ways for remote students to participate in class activities (e.g., “Google Jamboards for brainstorming if they want to instead of drawing a table on the white board.”). He shared that he generally

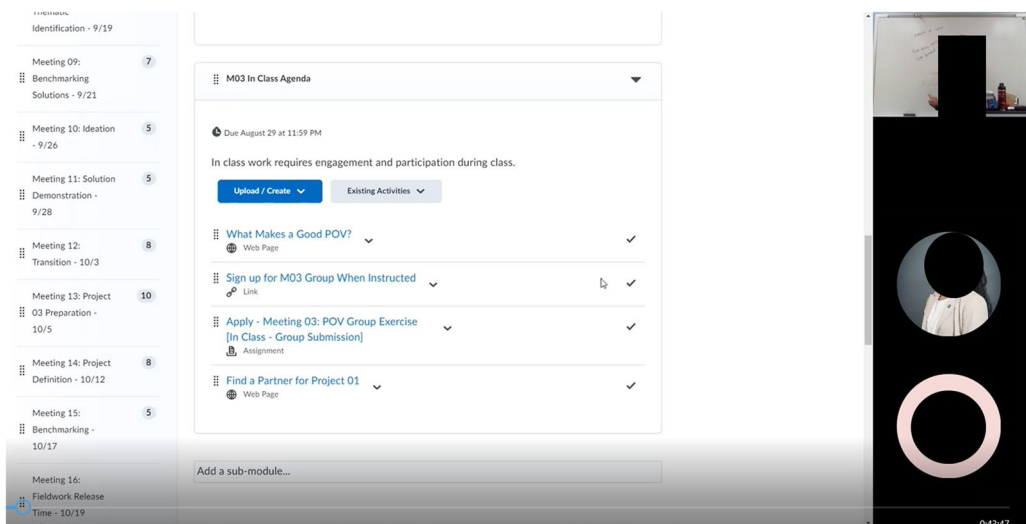


Figure 4. Example of Tom’s HyFlex setup.

tried to keep things the same across modalities and did not see “significant differences.” To promote students successfully navigating and self-regulating a HyFlex learning environment, Tom discussed best practices with students and had them practice using Teams.

Tom's social presence

Across class sessions, Tom was observed commonly using greetings or salutations with face-to-face students (e.g., “Have a good week and weekend. We’ll see you then.”) and disclosing personal information (e.g., sharing a picture of his family from a camping trip). However, in observed session he did not acknowledge remote students at the beginning of class and only used student names in a few instances. In a couple of instances, he used group language (e.g., “Anyway, today we’re going to take a look at the first two steps in our design process.”) and offered approval to students for their work (e.g., “I’ve seen some really good ones so far today, and I would expect nothing less.”). While he did invite responses from students, these focused on probing students for more information (e.g., “Can you just give us a brief recap?”) or checking-in (e.g., “So, are there any questions about this?”) versus inviting students to have a full-class discussion on a topic, which he explained during his interview that he tended to “shy away from” both full group discussions and using the chat tool in Teams. However, he shared that he encouraged his students to use the Teams chat for their team communications.

Tom consistently shared his screen to offer a visual of course content and assignment details. In one instance, to support students’ collaborative effort, he created an individual channel on Teams to help a group work with a remote teammate. At the end of the semester, Tom used his instructor station web cam to stream team presentations. However, since the teams were not sharing their screen, presentation content was not viewable. Additionally, he offered different collaborative modes for Teams based on their composition: “If you have a mixed attending group... [use] the Google Doc template that’s linked there in meeting three... or you can just use the whiteboard... I’d prefer if you use the whiteboard so I can walk around and see everything.” Although during his interview, Tom shared bringing remote students into class conversations and building community, these were not observed in class recordings.

Tom's teaching presence

During the first day of class, Tom discussed office hours and alternative availability to meet with students *via* Teams, adding that he wouldn’t respond to messages after 8:00 P.M. When facilitating class meetings, Tom used the beginning of each class to share resources (e.g., “Now, we have a composite character profile one-pager in Brightspace.”), offer reminders (e.g., “Do not forget that this is an individual submission before the end of the day today.”), and explain how to navigate an assignment and work in teams with a remote teammate, while sharing his screen to show documents or the LMS. When gearing up for a project, he discussed the process like this:

So, make sure you sign up for the correct group... Those of you joining us remotely will get you into a group with people who are working here face-to-face. If you have a student that is joining you remotely, what you’ll do is go over on the left-hand side of our general channel...

In a couple of instances, Tom used encouragement with students (e.g., discussing that being a freshman doesn’t limit what they are capable of). During one class session, Tom offered students tips for successfully completing college. Tom was rarely observed using facilitating discourse indicators. Finally, Tom used examples and resources to support his instruction (e.g., this company uses “composite character profiles to design different cruise experiences” or details of a design workshop he facilitated, while displaying pictures). At the beginning of the semester, he used the Teams chat as an open forum but as the semester progressed, he did not use it in this way. In a couple of instances, he offered formative feedback on students’ work, solicited feedback on how the course was going through a midterm evaluation form and informally in class, and displayed rubrics, while discussing requirements.

Comparing across cases

Across cases, instructors did several things: developing strategies for identifying and monitoring remote students; checking-in with teams and individual students both in face-to-face and remote modalities about their progress and needs; facilitating group collaboration; and leveraging technology affordances to create their presence. However, the instructors implemented each of these strategies in different ways. For instance, two instructors required all students to log into Microsoft Teams for each session and used the chat feature of Teams to check-in with students and offer feedback. The other instructor required students only to log into Teams if they had remote students in their group and tended to avoid using the chat tool. Additionally, one instructor made himself extensively available to students through the Microsoft Teams software platform—taking calls at late hours or during informal times, while the other two instructors did not offer similar availability. Finally, one instructor seemed to focus primarily on face-to-face activities (i.e., writing on a physical whiteboard, while using his web cam versus using a digital option), while the other two instructors used technology with all students to facilitate team activities (see Figure 5).

The first class appeared to set the tone for instructor presence: Key differences in instructor presence were observed in terms of how learners were addressed, the roles that instructors embraced, the roles that students were expected to embrace, expectations of participation across modalities, and the flow of communication (See Table 3). These differences were amplified by the technology usage both for students participating in the session and for those accessing recordings later. While Tom effectively used screen sharing to demonstrate LMS navigation, he saw PowerPoint as only a tool for sharing information and missed out on using it or a different tool to facilitate social presence (e.g., emphasis, collaboration, invitation) or teaching presence (e.g., reminders, prompts, feedback), like how Ben and Lia used the tool—potentially creating a presence that could reach students across modalities. Similarly, Lia missed out on opportunities to use screen sharing to model HyFlex navigation and create a more accessible class.

Regardless of these differences in presence, when comparing student CoI survey responses across sections, we found no significant differences. Table 4 shows the mean and standard deviation for the teaching and social presence scores for Ben, Lia, and Tom. Presence scores had a potential range of 1–5. Teaching presence scores were higher than social presence with means ranging from 4.3–4.6 across instructors and the three subconstructs. As a benchmark, a score of 4 represented agreement, and a score of 5 represented a strong agreement. Thus, mean scores for each instructor were situated between agree and strongly agree to statements such as “The instructor clearly communicated important course topics” and “The instructor helped to keep course participants engaged and participating in productive dialogue.”

We tested each of the three subconstructs (design and organization, facilitation, direct instruction) for significant difference across instructors using a one-way ANOVA. Slight differences across instructors were neither statistically nor practically significant. Social presence mean scores were about one-half point lower and ranged from 3.7–4.2 across the subconstructs and instructors, suggesting on average students agreed with statements such as “Getting to know other course participants gave me a sense of belonging in the course” and “I felt comfortable participating in the course discussions.” Each of the three subconstructs (affective expression, open communication, group cohesion) were tested for significant difference across instructors using a one-way ANOVA. As with teaching presence, slight differences across instructors were neither statistically nor practically significant.

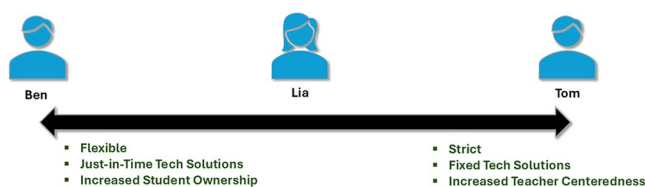


Figure 5. Continuum of instructor presence.

Table 3. Comparison of the first day of class.

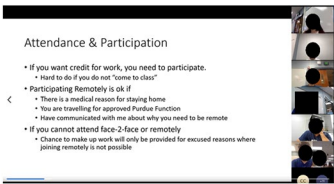
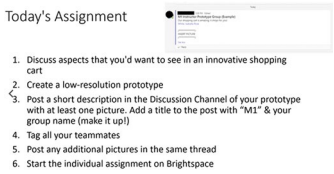
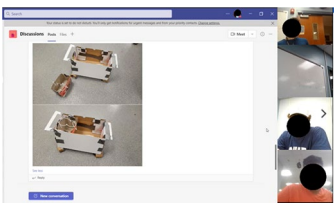
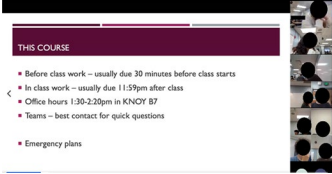
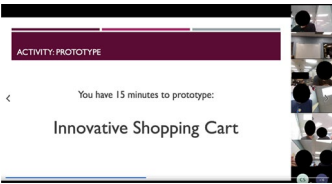
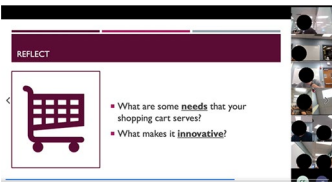
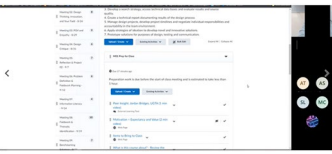
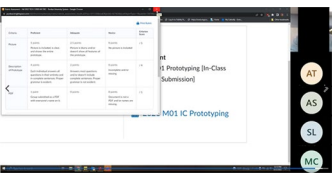
Ben	Lia	Tom
First 15 min of Class - Described professional background, introduced undergraduate teaching assistant (visible on video stream), explained HyFlex and attendance expectations and participation, offered tips for success in the class, and shared availability - Streamed video of himself speaking and used a presentation to emphasize points he was discussing, shared picture of himself, displayed prompts to facilitate the in-class activity, and provided example of prototyping, while also switching across Teams and LMS to demonstrate activities - Students were logged into the Teams meeting with videos on  Middle 20 min of Class - Provided five-minute overview of the prototyping activity - Facilitated students' working in groups to create prototypes - Created a discussion channel in Teams for students to share a picture of their prototype Today's Assignment - Discuss aspects that you'd want to see in an innovative shopping cart - Create a low-resolution prototype - Post a short description in the Discussion Channel of your prototype with at least one picture. Add a title to the post with "M1" & your group name (make it up!) - Tag all your teammates - Post any additional pictures in the same thread - Start the individual assignment on Brightspace  Last 15 min of Class - Student groups shared work with everyone, with Ben facilitating a discussion by asking them questions about their design - Pictures of the prototypes were displayed on the screen for viewing 	First 10 min of Class - Shared information about her professional and personal interests, emphasized the usefulness of design thinking, introduced TA (visible on video stream), described course policies (with help from TA), attendance, grading, and course structure, and shared reminders, deadlines, and availability - Streamed video of herself speaking and used a presentation to guide the things she discussed and question prompts for the class activity - Students were logged into the Teams meeting with videos on  Middle 20 min of Class - Provided three-minute overview of prototyping activity - Facilitated students' working in groups to create prototypes  Last 15 min of Class - Displayed reflection questions via presentation to debrief prototyping activity, asking student groups to share experience - Provided reminders to students about assignment 	First 20 min of Class - Discussed course structure and attendance, shared availability, described professional experience, introduced grader and TA (can hear TA but cannot see them on video stream), and displayed a picture of his baby - Streamed video of himself speaking, referred to information on dry erase board at times, and shared his screen as he displayed information on the LMS - Five students joined the Teams meeting with cameras off  Middle 15 min of Class - Provided three-minute overview of the prototyping activity, writing instructions on dry erase board (hard to see on video) - Facilitated students' working in groups to create prototypes, while displaying a clock with how much time they had to work in teams  Last 15 min of Class - Asked students to hold up their prototypes for everyone to see (less than 10 seconds) and provided a lecture-style debrief regarding design and prototyping (approximately 10 min) - Ended class with more demonstration on navigating Brightspace and information about grading 

Table 4. Descriptive statistics from CoI survey responses.

Presence	Ben Mean (SD)	Lia Mean (SD)	Tom Mean (SD)	<i>F</i> (2,111)	Sig
Teaching					
Design and Organization	4.5 (0.88)	4.4 (0.65)	4.5 (0.55)	0.167	.85
Facilitation	4.3 (0.99)	4.4 (0.58)	4.4 (0.62)	0.062	.94
Direct Instruction	4.6 (0.72)	4.4 (0.54)	4.3 (0.71)	1.277	.28
Social					
Affective Expression	3.7 (1.00)	4.0 (0.64)	4.1 (0.64)	2.311	.10
Open Communication	4.0 (0.77)	4.1 (0.61)	4.2 (0.61)	0.531	.59
Group Cohesion	3.7 (0.96)	4.1 (0.60)	4.1 (0.63)	1.415	.25

However, when comparing attendance data, clear patterns existed across sections. Ben's remote attendance varied considerably, ranging from 0 to 9 students per meeting, averaging about 4 remote attendees per class. Lia's attendance fluctuated between 0 to 9 students, averaging about 3 remote attendees, while Tom consistently maintained the lowest remote participation with numbers ranging from 0 to 3 students and averaging less than 1 remote attendee per meeting. This pattern suggested that Ben's section had both higher and more variable remote attendance, while students in Tom's sections more often selected in-person attendance.

Limitations

In this research, we considered how instructor presence is established in problem-centered HyFlex learning through instructor interviews and observations of instructor facilitation and compared students' perceptions and attendance across sections. Student perceptions beyond the CoI survey or performance within the course and across sections were not considered and could offer additional insight into the impact of instructor presence. Additionally, this research considered instructor presence from three cases in problem-centered, Interactive Synchronous HyFlex learning. Findings may not be applicable to other programs. Additional research can help provide insight into instructor presence in other HyFlex learning environments.

Discussion, implications, and conclusion

Overall, results demonstrate several ways that instructors established and maintained their presence through creating teaching presence and fostering social presence in HyFlex environments. Instructors appeared to operate on several continuums regarding technology use, attendance modality recommendations, and student autonomy. Designing and facilitating learning experiences is a complex process (Jonassen, 2008; Tawfik & Gatewood, 2022), and individual beliefs about teaching, learning, and technology can be influential factors on how educators design and facilitate learning (Ertmer, 2005; Ertmer et al., 2012). The differences in presence observed across instructors appeared to be rooted in how they conceptualized teaching and learning and viewed technology as an educational tool. The HyFlex setup in the course then amplified these differences.

When facilitating learning experiences, instructors can embrace many roles (Richardson et al., 2015; see Table 1). In HyFlex learning, these roles are necessary for establishing an effective climate and promoting meaningful learning experiences (Gascoigne, 2025). Instructors in this research used different roles to teach the same course—Tom predominately used organizing and maintaining; Lia focused on advocating and facilitating; and Ben mainly offered sense making and facilitating. However, across sections, students logged similar responses on the CoI survey. Although there were noticeable differences across how instructors established their presence, students seemed to adapt to and follow whatever the established norm was for the class. For instance, Tom created strict guidelines around when attending remotely was allowed and created an instructor-centered experience, while Ben was more flexible with students' remote attendance and transferred more ownership to students. Although the role of HyFlex was used very

differently across sections, students in each group appeared to understand this role and considered their instructors from the contexts the instructors established. Previous research (Glazier & Harris, 2021) suggests that students perceive several instructor strategies as meaningful (e.g., clear instructions; interactivity; instructor availability and promptness in responses). The presence established by the instructors in this course aligns with these actions, suggesting effective presence in HyFlex learning environments can be facilitated in multiple ways.

Furthermore, differences in the roles instructors embraced likely explain differences in attendance patterns. That is, although the course was designed using the CoI framework and the same HyFlex model, the facilitation and established presence across sections was different. Students in Tom's class appeared reluctant to use remote options to attend class, while students in Ben's class more frequently took advantage of remote options. Essentially, Tom's course operated as a face-to-face course, enhanced by screensharing and course recordings to support absences or further study, while Ben's course was implemented as flexible, meeting evolving needs. Similar to previous research (Gascoigne, 2025; Yang et al., 2025), findings from this study suggest the climate created in HyFlex environments can create norms and influence attendance. Generally, in problem-centered learning, instructors must guide students toward meaningful learning while transferring ownership in the learning process to students and how instructors take advantage of environmental technology affordances can impact the learning the overall learning experience (Ertmer & Koehler, 2018). A commonality across instructors was how each leveraged technology affordances to create presence. Key differences of technology use were observed across facilitating collaboration, offering availability, sharing information, and supporting remote students. As an example, when observing the same class session, Lia shared PowerPoint slides with the key points *via* Teams, while Tom projected the assignment *via* Teams. Ben used technology to facilitate discussion, support team collaboration, offer feedback, and suggest alternative digital collaboration tools for remote and in-person student collaboration. However, Lia and Tom did not specify any alternative collaboration tools for both modalities nor used technology to support deeper student-centered activities. In short, the way the instructors used technology appeared to align with how they viewed their role in the learning process, which in turn impacted the presence they created.

While suggesting that limiting the use of technology simultaneously limited the possible instructor presence seems plausible, the interaction of these elements is complex. For instance, while Tom was more hesitant about HyFlex participation and using technologies to support facilitation, his students primarily attended face-to-face. Therefore, an increase in technology use in Tom's case may not have enhanced his specific type of presence. However, instructors certainly missed opportunities for using technology to provide a more supportive, accessible, and student-centered learning experience, a potential benefit of HyFlex learning (Finkel-Gates, 2025). While research suggests that HyFlex in higher education is becoming the new normal for supporting students (Sellnow-Richmond et al., 2020) and students expect more flexibility in learning modality choice (Gascoigne, 2025), HyFlex also offers opportunities to facilitate meaningful *reflection* for instructors—an important tool for improving teaching and instructional design decision-making (Stefaniak et al., 2022). That is, the results of this research revealed opportunities simultaneously exist for developing and refining facilitation skills. Class recordings can be useful for instructors to reflect upon their own HyFlex presence. During interviews, instructors indicated strategies they were or were not using to facilitate HyFlex learning that did not match with observations, suggesting a disconnect between intention and execution or a lack of awareness of effective or ineffective strategies. Potentially, if instructors revisited videos similar to what we did in this research, they could identify ways they could enhance HyFlex learning for their students (e.g., looking into the camera to simulate eye contact and promote acknowledgement, using digital tools that bridge modalities and offer formative feedback) and creating more accessible learning. Finally, instructors had the benefit of being part of a community due to the large enrollment, multi-section nature of the course. As the roles embraced varied across instructors, they could learn best practices from one another and be inspired to implement new strategies and improve their approach. Creating opportunities for debriefing could offer meaningful opportunities for this sharing to occur.

Overall, this study offers insight into the design and facilitation of HyFlex learning environments, given their unique affordances. While previous studies have examined HyFlex learning from students' perspectives or performance, this study offers insight into the instructor perspective, extending understanding of how to effectively facilitate HyFlex learning experiences. Results suggest there is not one right answer, as three instructors, with different styles, were all able to establish an effective presence. For individuals interested in integrating HyFlex, the key is to envision the desired environment and think of HyFlex and the associated software as a toolset: Which learning experiences are best created by which tool(s) in your classroom? Additionally, when facilitating a HyFlex learning experience, communicating with the students how to be successful is essential, and they will interface with the system in the way it is created and communicated. While previous studies have examined HyFlex learning from students' perspectives or performance, this study offers insight into the instructor perspective, extending understanding of how to effectively facilitate HyFlex learning experiences.

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References

- Adam, M. S., Abd Hamid, J., Khatibi, A., & Azam, S. F. (2025). Investigating the effects of direct instruction and facilitated discourse on social and cognitive presence in blended learning. *Online Learning*, 29(1), 89–108. <https://doi.org/10.24059/olj.v29i1.4182>
- Adnan, M. (2017). Perceptions of senior-year ELT students for flipped classroom: A materials development course. *Computer Assisted Language Learning*, 30(3-4), 204–222. <https://doi.org/10.1080/09588221.2017.1301958>
- Amirova, A., Zhumabayeva, A., Zhunusbekova, A., Kalbergenova, S., Nygymanova, N., & Arenova, A. (2023). Effect of using Hyflex technology learning on preservice teachers' success and attitudes. *International Journal of Education in Mathematics, Science and Technology*, 11(3), 623–642. <https://doi.org/10.46328/ijemst.3309>
- Anderson, T., Rourke, L., Garrison, D. R., & Archer, W. (2001). Assessing teacher presence in a computer conferencing context. *Journal of Asynchronous Learning Networks*, 5(2), 1–17. <http://hdl.handle.net/2149/725>
- Arbaugh, J. B., Cleveland-Innes, M., Diaz, S. R., Garrison, D. R., Ice, P., Richardson, J. C., & Swan, K. P. (2008). Developing a community of inquiry instrument: Testing a measure of the Community of Inquiry framework

- using a multi-institutional sample. *The Internet and Higher Education*, 11(3-4), 133–136. <https://doi.org/10.1016/j.iheduc.2008.06.003>
- Beatty, B. J. (2007). Hybrid classes with flexible participation options—If you build it, how will they come. 2007 *Annual Proceedings-Anaheim*, 15.
- Beatty, B. J. (2019). *Beginnings: Where does hybrid-flexible come from*. Beatty, BJ hybrid- flexible course design: Implementing student-directed hybrid classes. EdTech Books.
- Boehm, M., & Boerboom, S. (2023). Faculty experiences of HyFlex: An exploratory study. *Educational Research: Theory and Practice*, 34(2), 43–47. <https://files.eric.ed.gov/fulltext/EJ1395229.pdf>
- Bozan, K., Gaskin, J., & Stoner, C. (2024). Student engagement in the HyFlex and online classrooms: Lessons from the COVID-19 Pandemic. *Technology, Knowledge and Learning*, 29(1), 509–536. <https://doi.org/10.1007/s10758-023-09661-x>
- Buckley, J. B., Thompson, A. K., Tretter, T. R., Biesecker, C., Robinson, B. S., & Hammond, A. N. (2024). “Where I feel the most connected”: Community of Inquiry supporting sense of belonging in a HyFlex engineering course. *The Internet and Higher Education*, 60, 100930. <https://doi.org/10.1016/j.iheduc.2023.100930>
- Cliff, S., & Assiouras, I. (2023). The transformation of post pandemic hybrid teaching and learning through experiences of remote digital learning in French business schools. *Innovations in Education and Teaching International*, 60(6), 810–823. <https://doi.org/10.1080/14703297.2022.2132982>
- Detyna, M., & Koch, M. (2023). An Overview of student perceptions of hybrid flexible learning at a London HEI. *Journal of Interactive Media in Education*, 20(4), 1–14. <https://doi.org/10.9743/JEO.2023.20.4.15>
- Edulee, N. B., Murphy, L., Emigh-Guy, M., & Croteau, K. (2023). Student perceptions about HyFlex/hybrid delivery of courses during the COVID-19 Pandemic. *College Teaching*, 72(4), 346–357. <https://doi.org/10.1080/87567555.2023.2208815>
- Ertmer, P. A., & Koehler, A. A. (2018). Facilitation strategies and problem space coverage: Comparing face-to-face and online case-based discussions. *Educational Technology Research and Development*, 66(3), 639–670.
- Ertmer, P. A. (2005). Teacher pedagogical beliefs: The final frontier in our quest for technology integration? *Educational Technology Research and Development*, 53(4), 25–39. <https://link.springer.com/article/10.1007/BF02504683> <https://doi.org/10.1007/BF02504683>
- Ertmer, P. A., Ottenbreit-Leftwich, A. T., Sadik, O., Sendurur, E., & Sendurur, P. (2012). Teacher beliefs and technology integration practices: A critical relationship. *Computers & Education*, 59(2), 423–435. <https://doi.org/10.1016/j.compedu.2012.02.001>
- Finkel-Gates, A. (2025). Embracing inclusivity and flexibility in higher education: A study of HyFlex delivery and its impact on diverse undergraduates in the post-pandemic era. *Open Scholarship of Teaching and Learning*, 4(1), 12–31. <https://osotl.org/osotl/article/view/113> <https://doi.org/10.56230/osotl.113>
- Garrison, D. R., Anderson, T., & Archer, W. (1999). Critical inquiry in a text-based environment: Computer conferencing in higher education. *The Internet and Higher Education*, 2(2-3), 87–105. [https://doi.org/10.1016/S1096-7516\(00\)00016-6](https://doi.org/10.1016/S1096-7516(00)00016-6)
- Gascoigne, C. (2025). Examining the effect of physical presence on perceived classroom climate in hyflex spaces. *Research on Education and Media*, 17(1), 49–55. <https://doi.org/10.2478/rem-2025-0006>
- Glazier, R. A., & Harris, H. S. (2021). Instructor presence and student satisfaction across modalities: Survey data on student preferences in online and on-campus courses. *International Review of Research in Open and Distributed Learning*, 22(3), 77–98. <https://doi.org/10.19173/irrodl.v22i3.5546>
- Guetterman, T. C., & Fetters, M. D. (2018). Two methodological approaches to the integration of mixed methods and case study designs: A systematic review. *American Behavioral Scientist*, 62(7), 900–918. <https://doi.org/10.1177/0002764218772641>
- Heilporn, G., & Lakhal, S. (2021). Converting a graduate-level course into a HyFlex modality: What are effective engagement strategies? *The International Journal of Management Education*, 19(1), 100454. <https://doi.org/10.1016/j.ijme.2021.100454>
- Joksimović, S., Gašević, D., Kovanović, V., Riecke, B. E., & Hatala, M. (2015). Social presence in online discussions as a process predictor of academic performance. *Journal of Computer Assisted Learning*, 31(6), 638–654. <https://doi.org/10.1111/jcal.12107>
- Jonassen, D. H. (2008). Instructional design as design problem solving: An iterative process. *Educational Technology*, 21–26. <https://www.jstor.org/stable/44429574>
- Jones, S. L., Worrall, M. J., & Woodards, K. L. (2024). Building a caring HyFlex pedagogy: An example of practice in an initial teacher education program. *Journal of Applied Learning and Teaching*, 7(1), 67–78. <https://doi.org/10.37074/jalt.2024.7.S1.8>
- Jorge, I. (2010). Social presence and cognitive presence in an online training program for teachers of Portuguese: Relation and methodological issues. In *IODL and ICEM 2010 joint conference and media days* (pp. 427–435) Publisher.
- Kohnke, L., & Moorhouse, B. L. (2021). Adopting HyFlex in higher education in response to COVID-19: Students’ perspectives. *Open Learning*, 36(3), 231–244. <https://doi.org/10.1080/02680513.2021.1906641>
- Magana, A. J., Karabiyik, T., Thomas, P., Jaiswal, A., Perera, V., & Dworkin, J. (2022). Teamwork facilitation and conflict resolution training in a HyFlex course during the COVID -19 pandemic. *Journal of Engineering Education*, 111(2), 446–473. <https://doi.org/10.1002/jee.20450>

- Martin, F., Wang, C., & Sadaf, A. (2018). Student perception of helpfulness of facilitation strategies that enhance instructor presence, connectedness, engagement and learning in online courses. *The Internet and Higher Education*, 37, 52–65. <https://doi.org/10.1016/j.iheduc.2018.01.003>
- Mentzer, N., & Mohandas, L. (2022). Student experiences in an interactive synchronous HyFlex design thinking course during COVID-19. *Interactive Learning Environments*, 32(5), 1–16. <https://doi.org/10.1080/10494820.2022.2124423>
- Mentzer, N. J., Isabell, T. M., & Mohandas, L. (2024). The impact of interactive synchronous HyFlex model on student academic performance in a large active learning introductory college design course. *Journal of Computing in Higher Education*, 36(3), 619–646.
- Mentzer, N., Mammadova, E., Koehler, A., Mohandas, L., & Farrington, S. (2025). Analyzing the impact of basic psychological needs on student academic performance: A comparison of post-pandemic interactive synchronous hyflex and pre-pandemic traditional face-to-face instruction. *Educational Technology Research and Development*, 73(1), 91–114.
- Merriam, S. B., & Tisdell, E. J. (2016). *Qualitative research: A guide to design and implementation* (4th ed.) Josey-Bass.
- Mohandas, L. (2022). *The impact of Interactive Synchronous HyFlex model on students' perception of social, teaching and cognitive presence in a design thinking course* (Publication No. 13123365) [Doctoral dissertation, Purdue University Graduate School]. ProQuest Dissertations Publishing.
- Nunes, K., Laliberté, N., & Rawle, F. (2023). The case for flexibility in online science courses: Strategies and caveats. *Biochemistry and Molecular Biology Education: A Bimonthly Publication of the International Union of Biochemistry and Molecular Biology*, 51(1), 89–93. <https://doi.org/10.1002/bmb.21690>
- Polat, H. (2022). Instructors' presence in instructional videos: A systematic review. *Education and Information Technologies*, 28(7), 1–33. <https://doi.org/10.1007/s10639-022-11532-4>
- Raes, A., Detienne, L., Windey, I., & Depaepe, F. (2020). A systematic literature review on synchronous hybrid learning: Gaps identified. *Learning Environments Research*, 23(3), 269–290. <https://doi.org/10.1007/s10984-019-09303-z>
- Reio, T. G., & Crim, S. J. (2013). Social presence and student satisfaction as predictors of online enrollment intent. *American Journal of Distance Education*, 27(2), 122–133. <https://doi.org/10.1080/08923647.2013.775801>
- Richardson, J. C., Koehler, A. A., Besser, E. D., Caskurlu, S., Lim, J., & Mueller, C. M. (2015). Conceptualizing and investigating instructor presence in online learning environments. *The International Review of Research in Open and Distributed Learning*, 16(3).
- Richardson, J. C., Besser, E. D., Koehler, A. A., Lim, J., & Strait, M. (2016). Instructors' perceptions of instructor presence in online learning environments. *The International Review of Research in Open and Distance Learning*, 17(4), 82–104.
- Richardson, J. C., Fiock, H., Olesova, L., & Stenbom, S. (2024). Evidence-based practices for developing a community of inquiry. *Handbook of research in online learning: Insights and advances*. Brill.
- Robinson, H., Kilgore, W., & Warren, S. (2017). Care, communication, support: Core for designing meaningful online collaborative learning. *Online Learning*, 21(4). <https://doi.org/10.24059/olj.v21i4.1240>
- Rosser-Majors, M. L., Rebeor, S., McMahon, C., Wilson, A., Stubbs, S. L., Harper, Y., & Sliwinski, L. (2022). Improving retention factors and student success online: Utilizing the Community of Inquiry Framework's instructor presence model. *Online Learning*, 26(2), 6–33. <https://doi.org/10.24059/olj.v26i2.2731>
- Sellnow-Richmond, D., Strawser, M. G., & Sellnow, D. D. (2020). Student perceptions of teaching effectiveness and learning achievement: A comparative examination of online and hybrid course delivery format. *Communication Teacher*, 34(3), 248–263. <https://doi.org/10.1080/17404622.2019.1673456>
- Shea, P., Richardson, J., & Swan, K. (2022). Building bridges to advance the community of inquiry framework for online learning. *Educational Psychologist*, 57(3), 148–161. <https://doi.org/10.1080/00461520.2022.2089989>
- Sheridan, K., & Kelly, M. A. (2010). The indicators of instructor presence that are important to students in online courses. *Journal of Online Learning and Teaching*, 6(4), 767–779. https://jolt.merlot.org/vol6no4/sheridan_1210.pdf
- Stefaniak, J., Baaki, J., & Stapleton, L. (2022). An exploration of conjecture strategies used by instructional design students to support design decision-making. *Educational Technology Research and Development*, 70(2), 585–613. <https://doi.org/10.1007/s11423-022-10092-1>
- Tawfik, A. A., & Gatewood, J. (2022). Decision making and problem-solving: Implications for learning design. *Journal of Applied Instructional Design*, 11(2), 7–23. <https://doi.org/10.51869/112/atjg>
- Tawfik, A. A., & Lilly, C. (2015). Using a flipped classroom approach to support problem-based learning. *Technology, Knowledge and Learning*, 20(3), 299–315. <https://doi.org/10.1007/s10758-015-9262-8>
- Velasquez, A., Graham, C. R., & Osguthorpe, R. (2013). Caring in a technology-mediated online high school context. *Distance Education*, 34(1), 97–118. <https://doi.org/10.1080/01587919.2013.770435>
- Vesely, P., Bloom, L., & Sherlock, J. (2007). Key elements of building online community: Comparing faculty and student perceptions. *MERLOT Journal of Online Learning and Teaching*, 3(3), 234–246. <https://jolt.merlot.org/vol3no3/vesely.pdf>
- Wong, B. T., Li, K. C., Chan, H. T., & Cheung, S. K. (2023). HyFlex learning research and practice: A longitudinal analysis. *Sustainability*, 15(12), 9699. <https://doi.org/10.3390/su15129699>
- Yang, H. H., Yin, Z., & Zhu, S. (2025). Examining students' acceptance of the large-scale HyFlex course: An empirical study. *British Journal of Educational Technology*, 56(1), 42–60. <https://doi.org/10.1111/bjet.13477>
- Yin, R. (2018). *Case study research and applications: Design and methods* (6th ed.). Sage Publications, Inc.

Appendix A

Teaching presence

Design & organization

1. What strategies did you use to communicate with students in the first classes of the semester related to TECH 120 using a HyFlex model?

-specifically related to communicating how the model relates to course topics, goals, and students' participation in learning activities?

Facilitation

2. How do you use the specific components of the HyFlex model to facilitate learning? After the initial response, follow-up with the prompts below—for instance, things like-

- keeping students on task
- prompting students to explore new topics
- focusing discussion on relevant issues

Direct instruction

3. How do you or plan to gauge learning and to provide feedback to students on their performance toward course goals in your HyFlex classroom? Can you share examples of how you've done this or plan to do this?

Social presence

Affective expression

4. How do you gain a sense of your students as real people and share your presence in your HyFlex classroom?

-If they don't talk about their own presence, prompt them to also consider how they use HyFlex components to create instructor presence?

Open communication

5. What strategies have you used or plan to use to make your HyFlex classroom a comfortable place for students to communicate and interact?

Group cohesion

7. What are specific ways that you manage group dynamics in your HyFlex classroom?

- disagreements
- collaboration

Cognitive presence

Triggering event

8. How do you pose problems for learners to consider in your HyFlex classroom? Can you provide any examples of how you've done this?

Exploration, integration, and resolution

9. How have you seen students use the HyFlex setting to explore and apply course content? Follow up with (if needed)– How did you intentionally facilitate this?

10. What approach do you use to prompt reflection in your HyFlex classroom? What does that look like?

Self-regulation

11. How do you support students' self-regulation in participating in a HyFlex learning experience?

Appendix B

Highlighted social presence			
	Tom	Ben	Lia
Affective	Indicator: Rich Media Activity: Checking with remote students that they are able to see and hear Quote: "I've gotten to start every class with, 'Hey! Are you able to hear and see me? OK.' Then, move on."	Indicator: Value Activity: Providing a course overview, while screensharing presentation Quote: "You'll learn a lot of concepts, but ultimately, practice is the best way to get good at design."	Indicator: Rich Media Activity: Starts class through video in Teams meeting, asking remote students to share <i>via</i> chat Quote: "...ask who's online so I can know who all is out there in the virtual setting... It's kind of hard to tell which background is the classroom..."
Cohesive	Indicator: Greetings and Salutations Activity: Welcoming remote students to the class session Quote: "If I see somebody's working remotely, I'll acknowledge them... I say, 'welcome everybody and welcome so and so joining us for remotely here today...we're glad that you're here.'"	Indicator: Collaborative Activity: Using Teams chat and video to facilitate group work Quote: "I like to have them comment on what projects they want to work on... I like to have some post where it's going to be a little more active interactive where they actually going to be talking to each other and so they're not just seeing a bunch of names, they're actually communicating with the people. Also encouraging people's cameras to be on."	Indicator: Collaborative Activity: Offering support to groups when they have a remote participant Quote: "For the in-class students, I make sure that they've started a meeting and that remote person has gotten into that meeting and ask kind of if they need any help with facilitating the group meeting... then also offering the speaker pucks that the students can use."
Open communication	Indicator: Acknowledge Activity: Checking in with groups and acknowledging their progress Quote: "It's sitting down with the groups and listening to what they have to say, and then asking questions ... just seeing what they're working on."	Indicator: Invitation Activity: Created a discussion channel on Teams Quote: "I have a channel just for discussion questions. And then I don't use it every day, but just asking them to discuss what they read and I ask him a couple questions like that."	Indicator: Invitations Activity: Using team channels for group progress check-ins Quote: "I can drop a message into each one of those channels to say. 'Hey! Reply to this, and let me and your teammates know where you are so far on this assignment' so that it's kind of giving me an idea if people are falling behind."
Highlighted teaching presence			
	Tom	Ben	Lia
Design and organization	Indicator: Navigation Activity: Explains to students how to facilitate group work with a remote teammate Quote: "If you have a group member that's working remotely, then your team needs to be on Microsoft Teams to do that... but then everybody else is working face-to-face."	Indicator: Availability Activity: Receiving video calls on Teams Quote: "I will try to take calls even at weird times if I'm available."	Indicator: Climate Activity: Using private channels on Teams Quote: "Even though I have access to it, they seem to be pretty comfortable knowing that it's just us, 8 people versus 38 people that all have access to see what they're working on."

Facilitating discourse	Indicator: Tips Activity: E-mailing suggestions to a student who has been participating remotely Quote: "I have seen... where students would just kind of drop off because they didn't feel part of the community... I would say have you ever thought about coming back to class, and they do. It seems to work out really well."	Indicator: Encourage Activity: Receiving video calls on Teams when teams are meeting Quote: "...just reassure them that what they were doing was good; give them some guidance from there."	Indicator: Tips Activity: Provides students with a team contract template Quote: "I give them a template to follow, and it does incorporate the Hyflex Model. It asks them how important it is that their teammates are there in person."
Direct instruction	Indicator: Demonstration Activity: Sharing his screen to go through requirements in LMS Quote: "If they want to join on Teams, they can see it on their computer as well... everything that's in Brightspace because that's all you really need in order to complete."	Indicator: Clarify Activity: Using Teams to clarify expectations Quote: "Two groups weren't working in the Google shared folder. I wanted them to, and so I just quickly added their team, added their channel and said 'hey, this is like you wanna use the channel.' I did that for both of the groups and they're able to get that pretty easily."	Indicator: Demonstration Activity: Walking through technology requirements and demonstrates procedures for connecting to Teams meeting Quote: "...making sure that they bring the right equipment to class, and then, once they are in the classroom, I ask all of them to have their teams turned on and I joined the meeting."
